

DeepAware AI: Business, Operations & Growth Take-Home

Market-size and conversion figures are working assumptions, labeled where they appear; pricing reflects OpenArm's current published figures. Project: openarm.dev.

1. Prioritization Note

I completed all six tasks, with the most depth on the GTM one-pager and the partnership proposal because those carry the highest strategic risk. I gave the outreach email and LinkedIn launch post extra polish because they're the artifacts most likely to face customers, researchers, and partners directly. If I'd been time-boxed to three, I'd have led with the GTM, the partnership proposal, and the job posting.

Assumptions, stated once:

- OpenArm 2.0 is an open-source robotic arm (open CAD, firmware, software stack) plus data collection infrastructure for robot learning: teleoperation, demonstration recording, dataset export, policy deployment. The arm and the data stack are one product.
- Open source is the adoption layer. Revenue comes from assembled and calibrated units, kits, support, and the data stack.
- Spec assumption: OpenArm 2.0 sits a tier above hobby-class kits (higher payload, rigid construction, torque-controlled CAN actuators, embedded Linux controller). The positioning below depends on this; verifying it is a week-one task in section 7.
- Early on 2.0, community from 1.0, no traction numbers I can cite. So I cite none. The targets below are commitments to test, which is different from claiming results.

2. GTM One-Pager: Research Labs & Universities

Target segment. Robot-learning labs publishing at ICRA, IROS, and CoRL, plus the courses they teach. From conference output, plausibly 800–1,500 manipulation-focused groups worldwide (assumption, sized in section 8). The PI decides; the lab manager files the purchase order; the grad student is how we get discovered. Startups and enterprise are real segments with a different motion (credit cards and SLAs, no purchase orders), deliberately sequenced second.

Key message. *From unboxing to recorded demonstrations in a day, on hardware you can open, modify, and cite.* The enemy is the **hardware bring-up tax**: "we should collect some teleop data" routinely becomes weeks of sourcing motors, debugging CAN buses, and tuning controllers before the first demonstration is recorded. We sell time-to-first-experiment, and reproducibility is the multiplier, since reviewers increasingly expect setups other labs can actually replicate.

Why us (the hard question, answered up front). A lab today picks between hobby kits, closed mid-tier arms, and proprietary research arms. Our column uses OpenArm's published specs and pricing:

	Hobby kit (SO-101 class)	Closed mid-tier (UFactory/Trossen class)	OpenArm 2.0	Proprietary (Franka class)
Price	\$100–\$500	\$1.5k–\$10k	~\$3.5k single, \$6.5k bimanual	\$20k–\$50k+
Payload / stiffness	Low	Mid	Mid-to-high	High
Torque control	No	Partial / varies	Yes (CAN)	Yes
Open firmware & CAD	Often partial	No	Fully open	No
Robot-learning data stack	Community-built	No	LeRobot-compatible	No
Replication by another lab	Easy, cheap	Possible, closed	Easy, exact revision citable	Expensive

Hobby kits are the right first robot, and we should say so publicly. Against the closed mid-tier, where the real fight is, we win on open firmware, the open data stack, and replication. If the 2.0 revision lands softer than the published specs here, the positioning changes, and I'd rather learn that in week one than after printing brochures.

Channels. GitHub and the project Discord, where the community already is. The Hugging Face LeRobot community, where much of the low-cost robot-learning world gathers and where we publish teleoperation datasets to stay visible (OpenArm already runs in LeRobot). Conference demos at CoRL, ICRA, and IROS; a live teleop demo is the strongest sales asset we have. Replication as a channel: every paper citing "OpenArm 2.0, firmware v2.1" is an ad in front of exactly our audience, so we make citing trivial (CITATION.cff, hardware DOI, a public publications list). Direct outreach to PIs and lab managers.

Three concrete actions in the next 30 days:

1. **Ship the university procurement kit:** quote template on letterhead, net-30 terms, W-9 on the website, vendor registration in common university systems, sole-source justification template. Boring, and it removes the most common way a university sale dies, which is paperwork friction.
2. **Open the pilot pipeline:** a named list of 30 labs from recent ICRA/IROS/CoRL papers whose setups match the arm; 10 personalized outreach emails (section 3); target two committed pilots, with pricing questions built into every conversation.
3. **Land the launch and open the first partnership.** Produce the 60-second teleop demo video that anchors the launch post and every conference booth after it, and open the RealSense partnership conversation (section 4).

Guardrails, briefly. Pricing: three tiers (kit near BOM, assembled and calibrated as the primary line, supported as the margin tier), with the assembled unit publicly priced around \$3,500 single to \$6,500 bimanual (lease from ~\$299/month), and the tier structure refined from pilot interviews. Ops gates sales: before scaling outreach we need component lead times, assembly hours per unit, max units per month, and the cost of a return, and sales targets stay at or under ship capacity. Year-one working targets: 10 pilots in two quarters, 25 paying academic accounts by month twelve, repeat purchase above 30%, time-to-first-experiment under one day. Kill condition: if pilot-to-paid conversion lands under roughly 20%, we stop scaling outreach until we know whether positioning or pricing is the problem.

3. University Partnership Outreach Email

Example recipient: Professor Chelsea Finn, whose Stanford IRIS lab's ALOHA line of work defined low-cost teleoperation for imitation learning. A real name and real work, chosen to show how this gets customized; the

actual send list comes from the 30-lab build in section 2, and every email gets this level of specificity or doesn't get sent.

Subject: Open-source arm for demonstration collection, seeking your lab's critique

Hi Professor Finn,

The ALOHA line of work made the case better than anyone: imitation learning moves fastest when the hardware is cheap enough to replicate and hack on. We've been building on that argument. OpenArm 2.0 is a fully open robotic arm (CAD, firmware, ROS 2 stack) with torque-controlled CAN motors, designed for LeRobot-compatible data workflows, so a lab goes from unboxing to recorded demonstrations in a day.

I'm writing because your group builds the kind of systems OpenArm is meant to support. If this is going to be useful to serious robot-learning labs, I'd want feedback from people who know where current hardware actually slows research down.

If it would be useful, I'll put two assembled units in the lab at pilot pricing (terms stated up front), with our engineers on call during setup. Honest feedback is the whole ask; the designs are open either way. And if a pilot ever turns into a purchase, we handle university procurement properly: formal quotes, purchase orders, net-30.

Happy to ship one to a student you nominate, or grab 15 minutes whenever convenient. If the answer is "we'd just build our own," that's genuinely useful to hear too.

Aidan Oliss · DeepAware AI · openarm.dev

4. Partnership Proposal: RealSense (Sensor Manufacturer)

Why RealSense, over the obvious names. I considered NVIDIA and Hugging Face. We're a YC S25 company already in NVIDIA's Inception program, so that relationship is live and worth deepening; but Inception is a startup track, and the bigger NVIDIA work (Isaac, Jetson) is a longer-horizon ecosystem play more than a partnership we'd land this quarter. Hugging Face / LeRobot matters too, but as a community and dataset channel (it's in the GTM channels above), since OpenArm already runs in LeRobot. RealSense fits the brief's "sensor manufacturer" example directly, and it's the partnership we could plausibly land first.

The technical fit is already proven; the partnership isn't. OpenArm's hardware already ships with a modular camera mount base built around RealSense cameras: a D435 as a chest camera and a D405 at the wrist, specifically to support consistent, reproducible data collection. So the integration isn't a hope, it's in the CAD. What doesn't exist yet is the commercial relationship, which is exactly what a partnership proposal should be about.

The timing is unusually good. RealSense (formerly part of Intel) completed its spinout in 2025 with \$50M in funding and a stated focus on AI, robotics, and computer vision; its depth cameras already sit in a large share of the world's mobile robots and humanoids, and it is actively building out its robotics go-to-market. A newly independent sensor company expanding its robotics GTM is the kind of partner that could plausibly take this call.

What each side gets:

- *RealSense gets:* a designed-in reference platform in the fast-growing robot-learning and physical-AI research segment; placement in front of exactly the labs and startups buying perception hardware; teleoperation datasets collected on their cameras that become public proof points.
- *DeepAware gets:* a validated, supported perception bundle so labs don't have to solve camera integration themselves, which feeds the time-to-first-experiment promise directly; a credible hardware partner's name on the reference design; preferred component pricing or co-marketing that makes the assembled bundle more attractive than a DIY build.

First three steps:

1. **Weeks 1–3: lock the validated reference bundle.** Formalize the already-proven D435/D405 setup as a documented, supported "OpenArm + RealSense" perception configuration, with calibration and drivers working out of the box. Most of this exists; the work is making it official and co-owned.
2. **Weeks 3–8: publish a joint reference dataset.** Collect and release a teleoperation dataset recorded on the OpenArm + RealSense bundle (to the LeRobot Hub, where our audience already is) with a reproducible reference policy. This is the artifact that proves the bundle and gives both companies something concrete to point to.
3. **Weeks 8–12: turn it into a commercial and co-marketing arrangement.** Preferred component pricing for assembled units, a co-branded reference-design page, and joint presence at a robotics venue (CoRL, ICRA, or IROS). The ask is specific and backed by a working bundle, not a logo swap.

Success at 90 days: a documented, supported OpenArm + RealSense bundle; one joint reference dataset live with a reproducible policy; and agreed terms (component pricing and co-marketing) for shipping the bundle. If RealSense passes, the same playbook runs with Orbbec or Luxonis; the validated bundle matters more than the specific vendor.

5. LinkedIn Launch Post: OpenArm 2.0

Note: the post leads with a 60-second teleop demo video; the text below is the caption.

Nobody gets into robot learning because they love crimping connectors.

But that's how most projects actually start: motors on backorder, a CAN bus that won't enumerate, controller gains tuned by folklore. Weeks disappear before the first demonstration is recorded. The field politely calls it "bring-up." We call it the tax everyone pays before the real research starts.

Today we're shipping **OpenArm 2.0**. One goal: open the box in the morning, record teleoperation data in the afternoon.

The whole thing is open. CAD, firmware, software. Fork it, swap the gripper, cite the exact revision in your paper so the next lab can replicate your setup instead of emailing you about it.

Under the hood: torque-controlled CAN motors, an embedded Linux board you can SSH into, ROS 2 native, and designed for LeRobot-compatible data workflows. No black boxes between your policy and the motors.

Buy it assembled and calibrated, or build it yourself from the repo for free. Same designs either way.

We made this for labs publishing at ICRA, IROS, and CoRL, for ML engineers crossing into robotics, and for founders who need arms on benches this month.

If you want pilot units, or you want to tell us what we got wrong (please do), the repo and my DMs are open.

openarm.dev

6. Job Posting: Hardware/Firmware Engineer

Full-time · Silicon Valley · competitive salary plus early-stage equity

DeepAware AI builds OpenArm, an open-source robotic arm for robot learning. Researchers use our hardware to collect teleoperation data and deploy learned policies, which means our firmware is part of other people's experiments. When a control loop jitters, someone's dataset is worse. That is the standard this code lives under.

You'd be one of the first engineering hires, owning the layer between learned policies and physical motors, in public, because our firmware is open source and our users read the diffs.

What you'll do

- Own the motor control firmware: CAN-based control of the joint actuators, torque/position/velocity modes, and the real-time loop that keeps teleoperation feeling immediate.
- Own the embedded Linux side: the board that runs the ROS 2 stack, device bring-up, the update path, and making "SSH into your arm" reliable enough to be a selling point.
- Own mechanical integration with the rest of the team: actuator mounting and joint assemblies, cable routing through moving joints, tolerance decisions that determine whether a community member's home-printed part fits, and design-for-assembly choices that decide how fast we can build units.
- Improve bring-up for everyone: calibration routines, self-tests, and error messages good enough that a first-year grad student fixes the problem without opening an issue.
- Iterate the electronics: schematic review, first-article debugging with a scope, and the judgment calls between "fix in firmware," "respin the board," and "change the bracket."
- Work in the open: public repo, public issues, community PRs. You'll talk to the researchers using your code, sometimes the day you ship it.

You're probably a fit if

- You've shipped embedded C/C++ on hardware that people other than you depended on: robotics, drones, industrial, automotive, anywhere reality pushes back.
- You've done motor control (FOC or similar) or hard real-time loops, and can reason about timing budgets out loud.
- You're comfortable on both sides of the wall: bare-metal/RTOS firmware and embedded Linux userspace. ROS 2 experience is a plus.
- You can read a schematic and a tolerance stack well enough to flag the problem before fab, even when someone else owns the CAD.
- The idea of strangers reading your firmware makes you write better firmware.

A specific degree, robot-learning experience, or having all of the above at once: none of that is required. Strong on firmware and curious about the rest is the profile.

To apply: send a short note about the hardest embedded or motor-control bug you've chased (what it was, how you cornered it), plus a link to anything public you've built. The note matters more than the resume.

7. 30-Day Onboarding Plan: Business, Operations & Growth Hire

Role chosen: this one, since it's the plan I'd actually run.

Who to meet in the first week: both founders, separately (the vision, and how they divide decisions); the firmware/hardware lead (the spec truth behind section 2's positioning); whoever assembles and ships arms (the capacity truth); two or three active community members from Discord (the unfiltered product review); and, as soon as one exists, the first pilot lab's PI or lab manager.

Week 1: Live the product, learn the factory. Assemble or shadow the setup of an OpenArm end to end, timing every step; I can't sell time-to-first-experiment without having lived it. Come out of the build conversations with four numbers: landed BOM cost, assembly-plus-calibration hours per unit, current maximum units per month, and component lead times. Verify the spec assumptions behind section 2. Read every GitHub issue, Discord thread, and inbound email from the last 90 days; tag by segment and pain.

Week 2: Make buying possible, and gate it. Ship the university procurement kit (section 2, action 1). Stand up a minimal CRM (a spreadsheet is fine) with funnel stages: repo visit, docs, quote request, purchase order, repeat purchase. With the founders, set a sales ceiling from the week-one capacity numbers, so we never sell what we can't ship.

Week 3: Open the pipeline. The 30-lab named list and 10 personalized pilot emails (section 2, action 2), with pricing questions built into every conversation. Open the RealSense partnership conversation (section 4). Draft launch assets, including the demo video, with whoever owns the 2.0 ship date.

Week 4: First pilots, first memo. Milestones: two pilot agreements verbally committed, one purchase order in motion, the partnership conversation live. Write the first monthly memo covering both halves of the job: funnel numbers (however small), what ten real conversations taught us about pricing and positioning, and the ops snapshot (on-time ship rate, capacity utilization, RMA activity). Set the operating rhythm: weekly pipeline review, monthly memo.

What a successful first month looks like: a lab can go from "interested" to "purchase order submitted" without a single improvised document; we have ten conversations' worth of pricing evidence; the RealSense conversation has a date on it; and we know exactly how many arms we can ship next month before promising them to anyone.

8. What I Would Expand With More Time

- **A measured competitive teardown.** Buy an SO-101-class kit and a closed mid-tier arm and time time-to-first-experiment on each, so the section 2 matrix moves from assumption to benchmark.
- **Market sizing done properly.** The 800–1,500 lab estimate deserves a real build: conference author affiliations, course adoption counts, and a startup-segment estimate on top.
- **A full willingness-to-pay study** beyond the questions embedded in pilot conversations: tier structure, support pricing, whether calibration-and-warranty separates.
- **The data flywheel, designed properly.** If many labs collect teleoperation data on identical hardware, aggregated datasets (opt-in, properly licensed) may eventually matter more than arm sales. The licensing and community-trust questions deserve real work before it becomes a revenue line.
- **The startup and enterprise motions:** inventory, card checkout, support SLAs, volume pricing, sequenced after capacity supports them.
- **International and distributor strategy:** procurement outside the US, CE marking, regional assembly partners.
- **Course-adoption program:** a professor who puts OpenArm in a syllabus creates a cohort of trained users every year.
- **Isaac Sim:** the second platform partnership, reusing the same asset pipeline.